

1D Source free,convection-diffusion problem using FVM with UDS

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Case study

Previously I tried solving the 1D Convection equation for a high peclet number($Pe > 2$) $Pe = 5$ using Finite Volume Method with Central differencing scheme and it fails to provide accurate results. This is due to Central differencing being bounded for $-2 \leq Pe \leq 2$ and also that CD* is not transportive, it does not take into account the direction of information transport as it places equal weights on the information provided by the neighbouring nodes. I decided to use Upwind differencing scheme because the Global peclet number ($Pe_L = 250$) and grid peclet number ($Pe = 5$) are both ($\gg 1$) which implies that convection is highly dominant both on the domain and at the cell level. Since Upwind scheme is transportive it could be a good match for this type of problem.

Problem

Consider the transport of a general fluid property ϕ by convection and diffusion on a 1D domain of length $L = 1$ on the x axis $0 \leq x \leq 1$.

Conditions on the boundaries are Dirichlet boundary conditions :

1. $\phi = 1 = \phi_0$ at $x = 0$

2. $\phi = 0 = \phi_1$ at $x = 1$

Convection velocity is u and is uniform.

1. fluid property $\rho = 1 \text{ kgm}^{-3}$

2. Material property $\Gamma = 0.1 \text{ kg/m} \cdot \text{s}$

The Governing Equation :

$$\frac{d}{dx}(\rho u \phi) = \frac{d}{dx}(\Gamma \frac{d\phi}{dx}) \quad (1)$$

The exact solution to the problem is this problem is :

$$\frac{\phi - \phi_0}{\phi_L - \phi_0} = \frac{e^{\frac{\rho u x}{\Gamma}} - 1}{e^{\frac{\rho u L}{\Gamma}} - 1}$$

The Péclet number (Pe) compares the convective force to diffusive force.

- Pe $\ll$ 1: diffusion dominates
- Pe $\approx$ 1: convection and diffusion are comparable
- Pe $\gg$ 1: convection dominates

Write a python code that solves for convection diffusion problem for N = 50, and Pe = 5 .

Discretization

Cell centered FVM is used to construct 1D grid of $N = 50$ Nodes/Control volumes. Grid spacing $h = \frac{L}{N}$

Integrate the governing equation (1) onto the finite volume centered at P.

$$\int \frac{d}{dx}(\rho u \phi) dV = \int \frac{d}{dx}(\Gamma \frac{d\phi}{dx}) dV$$

Using gauss divergence theorem :

$$\int \hat{n} \cdot (\rho u \phi) dA = \int \hat{n} (\Gamma \frac{d\phi}{dx}) dV$$

Resulting in the flux equation :

$$\rho u \phi A)_e - \rho u \phi)_w = \Gamma A \left(\frac{d\phi}{dx} \right)_e - \Gamma A \left(\frac{d\phi}{dx} \right)_w$$

Given ρ cst., Γ cst , u cst. , h cst. , A cst.

We get :

$$\rho u (\phi)_e - \phi)_w = \Gamma \left(\frac{d\phi}{dx} \right)_e - \frac{d\phi}{dx} \right)_w \quad (2)$$

Let's assume that ϕ varies piecewise linear between nodes , we Choose Upwind differencing Scheme as discussed previously.

$$\phi_e = \begin{cases} \phi_P, & u > 0, \\ \phi_E, & u < 0. \end{cases}$$

$$\phi_e = \phi_P$$

$$\phi_w = \begin{cases} \phi_W, & u > 0, \\ \phi_P, & u < 0. \end{cases}$$

$$\phi_w = \phi_W$$

$$\left(\frac{d\phi}{dx} \right)_e = \frac{\phi_E - \phi_P}{h}$$

$$\left(\frac{d\phi}{dx} \right)_w = \frac{\phi_P - \phi_W}{h}$$

Using EQN (2) flux eqn :

$$\rho u (\phi_P - \phi_W) = \Gamma \left(\frac{\phi_E - \phi_P}{h} - \frac{\phi_P - \phi_W}{h} \right)$$

$$Pe(\phi_P - \phi_W) = \phi_E - \phi_P - \phi_P + \phi_W$$

$$(2 + Pe)\phi_P = \phi_E + (1 + Pe)\phi_W$$

$$a_P \phi_P = \phi_E + (1 + Pe)\phi_W \quad (A)$$

with :

$$a_P = a_E + a_W$$

$$a_E = 1,$$

$$a_W = 1 + Pe$$

Result (A) and its coefficients FOR ALL INTERIOR NODES

FOR NODE 1 : Boundary node , $\phi_0 = 1$ at $x = 0$

Using the flux eqn. (6)

$$\phi_e = \phi_P$$

$$\phi_w = \phi_0 = 1$$

$$\left(\frac{d\phi}{dx}\right)_e = \frac{\phi_E - \phi_P}{h}$$

$$\left(\frac{d\phi}{dx}\right)_w = \frac{\phi_P - \phi_0}{\frac{h}{2}} = 2\left(\frac{\phi_P - 1}{h}\right)$$

$$\rho u(\phi_P - 1) = \Gamma\left(\frac{\phi_E - \phi_P}{h} - 2\left(\frac{\phi_P - 1}{h}\right)\right)$$

$$(3 + Pe)\phi_P = \phi_E + 2 + Pe$$

$$a_P\phi_P = a_E\phi_E + (2 + Pe) \quad (B)$$

with :

$$a_E = 1, a_P = a_E + (2 + Pe)$$

Result (B) and its coefficients FOR Node 1

FOR LAST NODE : Boundary node , $\phi_L = 0$ at $x = L$

Using the flux eqn. (6)

$$\phi_e = \phi_P$$

$$\phi_w = \phi_W$$

$$\left(\frac{d\phi}{dx}\right)_e = \frac{\phi_L - \phi_P}{\frac{h}{2}} = 2\left(\frac{-\phi_P}{h}\right)$$

$$\left(\frac{d\phi}{dx}\right)_w = \frac{\phi_P - \phi_W}{h}$$

$$\rho u(\phi_P - \phi_W) = \Gamma(\frac{-2\phi_P}{h} - (\frac{\phi_P - \phi_W}{h}))$$

$$(3 + Pe)\phi_P = (1 + Pe)\phi_W$$

$$a_P\phi_P = a_W\phi_W \quad (C)$$

with :

$$a_W = 1 + Pe, a_P = a_W + 2$$

Result (C) and its coefficients FOR Last Node

Numerical implementation

- For interior nodes : $\phi_{new}[i] = (a_E\phi[i + 1] + a_W\phi[i - 1])/a_P, a_P = a_E + a_W, a_E = 1, a_W = 1 + Pe$
- For Node 1 : $\phi_{new}[i] = (a_E\phi[i + 1] + (2 + Pe))/a_P, a_E = 1, a_P = a_E + (2 + Pe)$
- For last Node : $\phi_{new}[i] = (a_W\phi[i - 1])/a_P, a_W = 1 + Pe, a_P = a_W + 2$

Numerical Results

For $N = 50$ and $Pe = 5$ and using FVM with UDS:

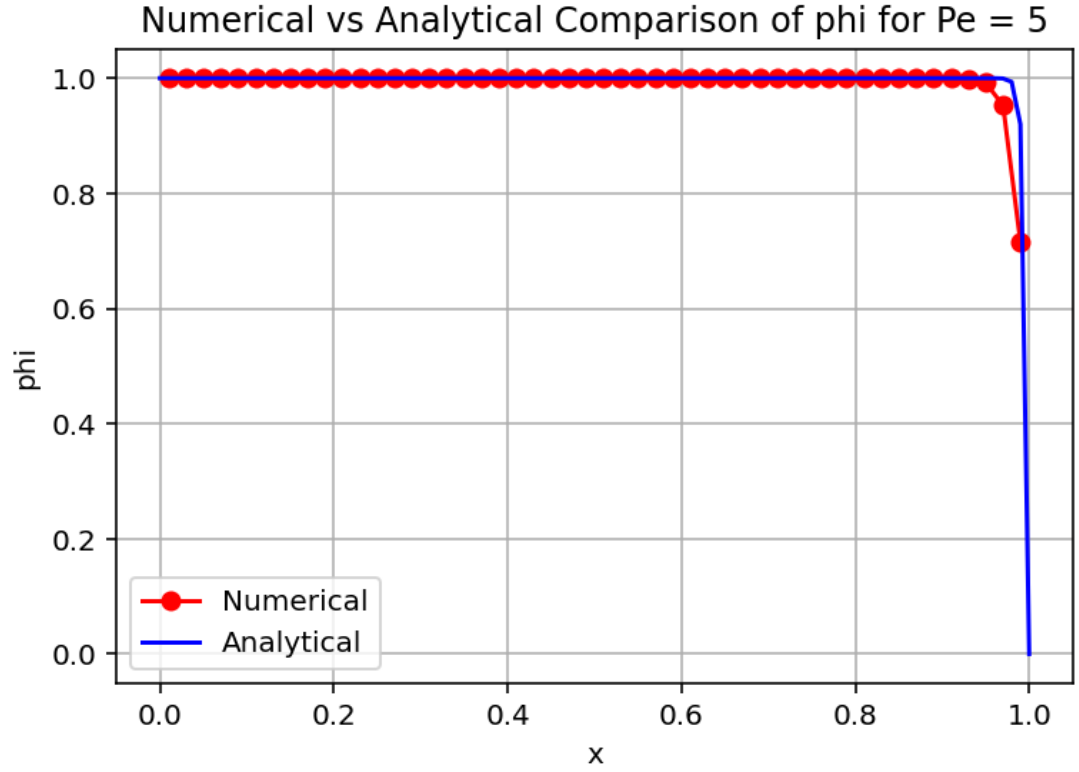


Figure 1: Plot of Numerical/analytical value of ϕ Vs x for $Pe = 5$ using UDS*

Table illustrates a comparison between Numerical and Analytical value of ϕ for $Pe = 5$

Node	Distance	Numerical solution of phi	Analytical solution of phi
1	0.01	1.0	1.0
2	0.03	1.0	1.0
3	0.05	1.0	1.0
4	0.07	1.0	1.0
5	0.09	1.0	1.0
6	0.11	1.0	1.0
7	0.13	1.0	1.0
8	0.15	1.0	1.0
9	0.17	1.0	1.0
10	0.19	1.0	1.0
11	0.21	1.0	1.0
12	0.23	1.0	1.0
13	0.25	1.0	1.0
14	0.27	1.0	1.0
15	0.29	1.0	1.0
16	0.31	1.0	1.0
17	0.33	1.0	1.0
18	0.35000000000000003	1.0	1.0
19	0.37	1.0	1.0
20	0.39	1.0	1.0
21	0.41000000000000003	1.0	1.0
22	0.43	1.0	1.0
23	0.45	0.9999999999999999	1.0
24	0.47000000000000003	0.9999999999999994	1.0
25	0.49	0.9999999999999991	1.0
26	0.51	0.9999999999999973	1.0
27	0.53	0.9999999999999959	1.0
28	0.55	0.9999999999999879	1.0
29	0.57000000000000001	0.9999999999999822	1.0
30	0.59	0.9999999999999484	1.0
31	0.61	0.9999999999999247	1.0
32	0.63	0.9999999999997834	1.0
33	0.65	0.9999999999996773	1.0
34	0.67	0.9999999999990395	1.0
35	0.69000000000000001	0.9999999999981782	1.0
36	0.71	0.9999999999930104	1.0
37	0.73	0.9999999999734587	1.0
38	0.75	0.9999999998561488	1.0
39	0.77	0.9999999991951111	1.0
40	0.79	0.9999999952288838	1.0
41	0.81	0.9999999715863559	1.0
42	0.83000000000000001	0.9999998297311882	1.0
43	0.85	0.9999989791412435	0.9999999999999999
44	0.87	0.9999938756015754	0.9999999999999862
45	0.89	0.9999632561838976	0.9999999999980447
46	0.91	0.9997795396778307	0.9999999997257663
47	0.93	0.9986772464453928	0.9999999620288001
48	0.95000000000000001	0.9920634870507646	0.9999948277163905
49	0.97	0.9523809470126655	0.999310467197038
50	0.99	0.7142857067840709	0.9107622823924667

Table 1: My data for Numerical value of phi for $Pe = 5$

For $N = 50$ and $Pe = 5$ and using FVM with Central differencing scheme:

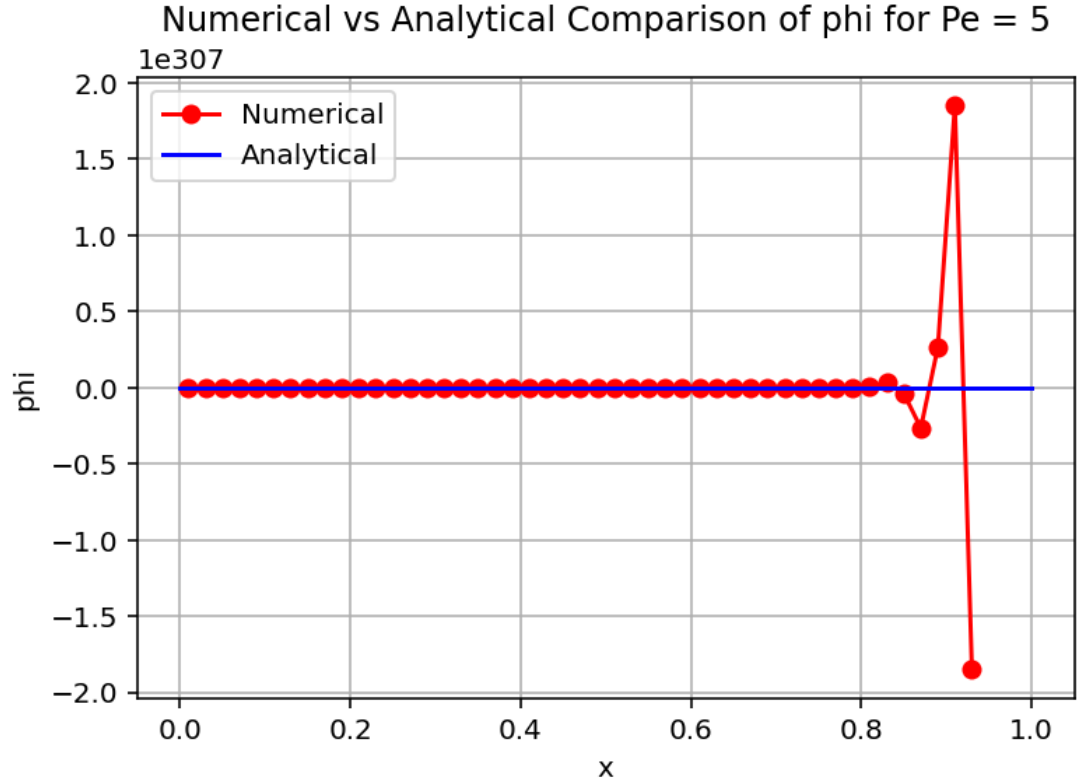


Figure 2: Plot of Numerical/analytical value of phi Vs x for $Pe = 5$ *Picture come from PROJECT 4

Discussion of Results

Consider the Global Péclet number $Pe_L = \frac{PeL}{h}$. For case $Pe = 5$ we have a $Pe_L = 250$. This case presents a global Péclet number which is $\gg 1$ which shows that convection force is way stronger than diffusion force over the whole domain and across a mesh. Physically we expect the property ϕ (initial value upstream) to be transported over a large distance before diffusion significantly changes it because convection transports ϕ faster than diffusion can spread it.

NOTE : CONVECTION carries/drag the value , Diffusion spreads the value

Diffusion starts to spread the value of ϕ after $x = 0.9$

Figure 1 shows that Upwind Differencing numerical scheme produces a solution which is bounded , therefore we can say that the scheme is bounded.

Figure 2 shows that Central Differencing numerical scheme produces a solution which is not bounded , therefore we can say that the scheme does not respect boundedness.

Boundedness says that in the absence of a source , the value of ϕ at internal nodes should be bounded by min and max of boundary values i.e 0 and 1

Conclusion

From this investigation we can say that Upwind difference scheme in Finite Volume Method produces far better results than Central differencing scheme for $Pe > 2$.